Institution or Laboratory Name

Course or Research Group

Electrical Characterisation of a Resistor

Laboratory Report

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Abstract

This short example demonstrates how to format a scientific laboratory report. A simple resistor is used as a familiar illustrative system. All values, observations and uncertainties are **synthetic** and do not describe an actual experiment. Replace this paragraph with the purpose, methods, central findings and conclusion of your own work.

1 Introduction

A laboratory report should explain the scientific question before presenting measurements. For an ideal ohmic resistor, voltage V and current I are related by

$$V = IR, \tag{1}$$

where R is the electrical resistance. Equation (1) is used only to illustrate mathematical typesetting and cross-references.

2 Experimental Methods

Describe the instruments, sample preparation and measurement sequence in enough detail to reproduce the work. Here, imagine a variable voltage source and a current meter connected to a nominal $1\,\mathrm{k\Omega}$ resistor. Assume example standard uncertainties of $0.01\,\mathrm{V}$ and $0.03\,\mathrm{mA}$, respectively. These are arbitrary teaching values.

Use consistent units and explain how uncertainties are estimated. A standard introduction to uncertainty analysis is given by Taylor [1997].

3 Results

Table 1 contains fictional observations. Numbers in an actual report should be reported with sensible significant figures and should be supported by the original measurement records.

Table 1: Synthetic voltage-current values, for layout demonstration only.

Voltage V (V)	Current I (mA)	Uncertainty $u(I)$ (mA)
0.50	0.50	0.03
1.00	1.01	0.03
1.50	1.49	0.03
2.00	2.02	0.03
2.50	2.49	0.03
3.00	3.01	0.03

Figure 1 illustrates a compact plot generated directly in LaTeX. The solid line is an ideal reference curve, not a fit to real data.

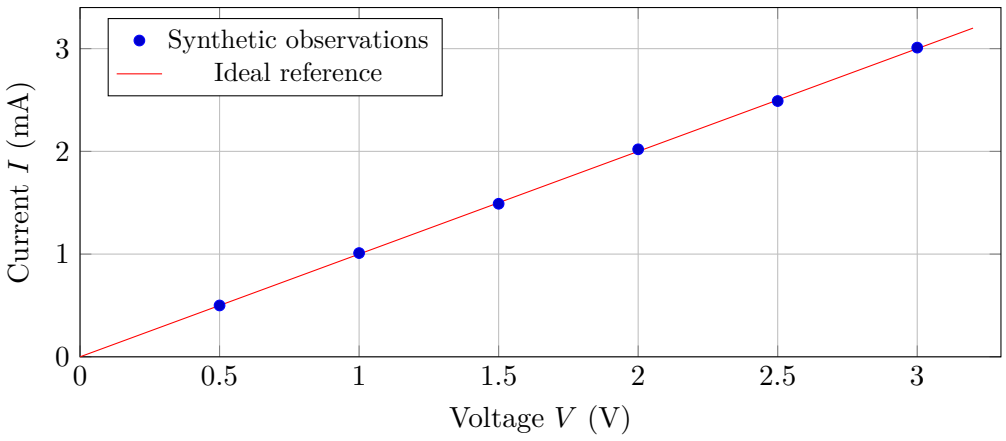


Figure 1: Illustrative current-voltage relationship. The data are synthetic.

4 Discussion

Discuss the physical interpretation rather than simply restating the results. For example, a resistance could be calculated from the slope of current against voltage. Under an independence assumption, illustrative propagation for $R = V/I$ is

$$\frac{u(R)}{R} \approx \sqrt{\left(\frac{u(V)}{V}\right)^2 + \left(\frac{u(I)}{I}\right)^2}. \tag{2}$$

Explain whether the assumptions behind Equation (2) are appropriate for the real experiment and distinguish random from systematic effects.

5 Conclusion

State what was established, what remains uncertain and what might improve a future measurement. Delete all fictional example content when adapting this template for an actual laboratory report.

References

John R. Taylor. *An Introduction to Error Analysis: The Study of Uncertainties in Physical Measurements*. University Science Books, 2 edition, 1997.